The empty set exists in Z if any set exists

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:emptyexists of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the Separation axiom schema of Z
- the extensional characterization of the empty set

Proposition 0.1. *If any set exists, then $\emptyset$ exists.*

Proof. If A is a set, $\emptyset = \{ x \in A \mid x \neq x \}$ exists by Separation. □

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